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Research article

Numerical and experimental study on spatter formation mechanisms and suppression method during ultra-high power laser-arc hybrid welding



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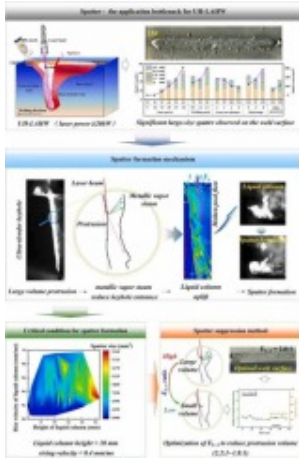
Highlights

- Ultra-high power laser-arc hybrid welding is employed for welding thick steel plates.
- Keyhole wall protrusions dictate keyhole-molten pool dynamics and spatter generation.
- Reducing protrusion volume mitigates liquid column ejection and spatter generation.
- Regulating laser-to-arc energy ratio threshold achieves effective spatter suppression.

Abstract

Ultra-high power laser-arc hybrid welding (UH-LAHW, laser power ≥ 20 kW) provides an effective method for joining thick-walled components. However, under ultra-high power laser irradiation, intense keyhole-molten pool fluctuations lead to spatter formation, which remains a critical challenge hindering the industrial application of UH-LAHW. This study systematically investigates spatter formation and suppression mechanisms in UH-LAHW through combined experimental and simulation approaches. The intrinsic relationships among keyhole wall protrusion dynamics, molten pool oscillations, and spatter formation were analyzed. Results indicate that front keyhole wall (FKW) protrusions exhibit two distinct motion modes under laser irradiation: “melt-evaporation-displacement” and “melt-evaporation-evaporation”. The former is identified as the dominant mechanism driving rapid upward movement of liquid columns and spatter formation. Regulating the laser-to-arc energy ratio threshold ($E_{L:A} = 2.2:1 - 1.8:1$) can reduce FKW protrusion volume, expand the keyhole opening, and decrease liquid column ejection height. This process culminated in the effective suppression of spatter at the optimal welding condition ($E_{L:A} = 2.0:1$). This study presents an effective approach for spatter suppression and provides a theoretical foundation and data support for industrial application of UH-LAHW technology.

Graphical Abstract



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Keywords

Ultra-high power laser-arc hybrid welding; Front keyhole wall; Protrusion structure; Liquid column; Spatter formation; Laser-to-arc energy ratio threshold

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